# AI/ML Engineering Roadmap

*A Complete Self-Study Path from Foundations to Professional AI Engineering*

## Study Order

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|---|---|
| **Phase 0** | Math Foundations + Python (run in parallel throughout) |
| **Phase 1** | Python · Statistics · EDA · Visualization (GFG Weeks 0–4) |
| **Phase 2** | SQL + Databases |
| **Phase 3** | Machine Learning + WQU Data Science Projects (GFG Weeks 5–10) |
| **Phase 4** | Git · Docker · CI/CD · Core MLOps |
| **Phase 5** | Deep Learning+ WQU DL Projects + Reinforcement Learning + DSA (GFG Weeks 11–16) |
| **Phase 6** | Modern AI Engineering (NLP · CV · LLMs · GenAI · RAG · Vector DBs · Agents) |
| **Phase 7** | Advanced Track — Cloud · Data Engineering · Google Professional ML Engineer |

# PHASE 0 Python + Math Foundations

Run Python (0.0) alongside every phase. Complete Math (0.1–0.4) before starting Phase 1.

## 0.0 — Python

→ **Python Programming MOOC 2026** — *University of Helsinki · free · world-class*
https://programming-26.mooc.fi/

## 0.1 — Calculus

→ **AP Calculus AB** — *Khan Academy*
https://www.khanacademy.org/math/ap-calculus-ab

## 0.2 — Linear Algebra

→ **Linear Algebra** — *Khan Academy*
https://www.khanacademy.org/math/linear-algebra

## 0.3 — Statistics & Probability

→ **Statistics & Probability** — *Khan Academy*
https://www.khanacademy.org/math/statistics-probability

## Alternative Path — Mathematics for Machine Learning (if Khan Academy feels too disconnected from AI)

**Mathematics for Machine Learning Specialisation** — *Imperial College London · Coursera · audit free · Linear Algebra, Multivariate Calculus & PCA*

## 0.1 — Calculus

→ **Multivariate Calculus** — *Imperial College London*
https://www.youtube.com/watch?v=cWZLPv4ZJhE&list=PLiiljHvN6z193BBzS0Ln8NnqQmzimTW23

→ **Essence of Calculus** — *3Blue1Brown · watch first*
https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr

## 0.2 — Linear Algebra

→ **Linear Algebra for Neural Networks** — *Imperial College London*
https://www.youtube.com/playlist?list=PLiiljHvN6z1_o1ztXTKWPrShrMrBLo5P3

→ **Essence of Linear Algebra** — *3Blue1Brown · must watch*
https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

### — PCA (after linear algebra)

→ **PCA Course** — *Marc Peter Deisenroth · co-author of Mathematics for Machine Learning*
*https://www.youtube.com/playlist?list=PL93aLKqThq4iDbY_wxfh3WgxkkzqH4aoo*

→ **Companion Textbook (free PDF)** — *mml-book.com*
https://mml-book.com

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## PHASE 1 Python · Statistics · EDA · Visualization

✅ Fully covered by GFG Weeks 0–4. No external course required.

**1.1 Data Science Overview** — *GFG Week 0*
**1.2 Python Foundations** — *GFG Week 1*
**1.3 Statistics for Data Science** — *GFG Week 2*
**1.4 EDA with Pandas & NumPy** — *GFG Week 3*
**1.5 Data Visualization — Matplotlib, Seaborn, Plotly** — *GFG Week 4*

**GFG Course: https://www.geeksforgeeks.org/batch/ds-16?tab=Resources**

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## PHASE 2 SQL + Databases

⚠ GFG does not teach SQL. Complete the resources below before Phase 3.

**2.1 SQL Murder Mystery** — *Joon Park and Cathy He*

https://www.kaggle.com/learn/intro-to-sql

**2.2 SQL Tutorial 1** — *Mode Analytics · primary*
https://mode.com/sql-tutorial/

**2.3 SQL Tutorial 2** — *SQL Zoo*
https://sqlzoo.net/wiki/SQL_Tutorial

**2.4 PostgreSQL Basics** — *YouTube · optional but recommended*
https://www.youtube.com/watch?v=qusjTv8N1cA

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## PHASE 3 Machine Learning + WQU DS Projects

✅ Fully covered by GFG Weeks 5–10. No external course required.

**3.1 ML Basics + Regression** — *GFG Week 5*

**3.2 Model Evaluation + SHAP / Model Explainability + Classification** — *GFG Week 6 + SHAP*

*Do SHAP / Model Explainability immediately after Day 36. It takes one afternoon and you'll use it in every project after.*

**3.2.1 SHAP / Model Explainability** — https://shap.readthedocs.io/en/latest/

**3.3 ML Algorithms — Decision Trees, SVM, KNN, Naive Bayes** — *GFG Week 7*

**3.4 Ensemble Learning — XGBoost, LightGBM, CatBoost** — *GFG Week 8*

**3.5 Unsupervised Learning — Clustering, GMM, DBSCAN** — *GFG Week 9*

**3.6 Dimensionality Reduction + Deployment Introduction** — *GFG Week 10*

### 3.7 — WQU Data Science Projects

→ **Applied Data Science Lab**
https://www.wqu.edu/adsl

## PHASE 4 Git · Docker · CI/CD · Core MLOps

⚠ GFG does not teach these tools. Complete before moving to Deep Learning.

### 4.1 — Git & GitHub

→ **Git & GitHub Full Course** — *freeCodeCamp*
https://www.freecodecamp.org/news/git-and-github-crash-course-for-beginners/

→ **Microsoft Learn for GitHub** — *Microsoft*
https://learn.microsoft.com/en-us/training/github/

### 4.2 — Docker

→ **Docker Tutorial for Beginners** — *TechWorld with Nana · YouTube*
https://www.youtube.com/watch?v=3c-iBn73dDE

→ **Docker for Beginners** — *freeCodeCamp*
https://www.freecodecamp.org/news/an-introduction-to-docker-and-containers-for-beginners/

### 4.3 — CI/CD

→ **GitHub Actions Full Course** — *YouTube*
https://www.youtube.com/watch?v=mFFXuXjVgkU

### 4.4 — Core MLOps

→ **Made With ML — MLOps** — *madewithml.com · best free MLOps resource available · primary*
https://madewithml.com

→ **MLOps Zoomcamp** — *DataTalksClub · alternative / highly practical*
https://datatalks.club/blog/mlops-zoomcamp.html

## PHASE 5 Deep Learning + WQU DL Projects + Reinforcement Learning + DSA

✅ Fully covered by GFG Weeks 11–16. No external course required.

**5.1 Introduction to Deep Learning** — *GFG Week 11*
**5.2 Optimisation + MLP + Backpropagation** — *GFG Week 12*
**5.3 CNN + RNN + Time Series** — *GFG Week 13*
**5.4 LSTM, GRU, Transformers** — *GFG Week 14*
**5.5 GANs + Autoencoders** — *GFG Week 15*

✅ GFG backup

**5.6 Practical Deep Learning** ***fast.ai***
**https://course.fast.ai/**

5.5 Dive into Deep Learning
https://d2l.ai/

## 5.6 — WQU Deep Learning Projects

→ **Deep Learning Fundamentals Lab**
https://www.wqu.edu/deep-learning-lab

## 5.7— Core Reinforcement Learning

5.7.1 Reinforcement Learning — *GFG Week 16*

**5.7.2 Reinforcement Learning** — ***Hugging Face Deep RL Course***
**https://huggingface.co/learn/deep-rl-course/unit0/introduction**

**5.7.2 PPO (Proximal Policy Optimization)** — ***Hugging Face Deep RL Course***
***https://huggingface.co/learn/deep-rl-course/unit8/introduction***

## 5.8 — DSA

5.8.1 Problem Solving with Algorithms and Data Structures using Python — *Runestone Academy*
https://runestone.academy/ns/books/published/pythonds/index.html

5.8.2 DSA *(Video lectures)* — *freeCodeCamp*
https://www.youtube.com/watch?v=pkYVOmU3MgA

5.8.3 Practice Problems — *Striver's A2Z Sheet*
https://takeuforward.org/dsa/strivers-a2z-sheet-learn-dsa-a-to-z

# PHASE 6 Modern AI Engineering

📖 **Companion Document**

Phase 6 has a dedicated full course guide covering all topics below in depth, with the top 3 free comprehensive courses per subject, structured in the exact order you should study them.

→ **See: Modern AI Engineering — Full Course Guide**

### Topics — in learning order

**6.1 Natural Language Processing** — *Transformers, BERT, GPT, text classification, NER, summarisation*
**6.2 Computer Vision** — *CNNs, object detection, segmentation, OpenCV, real-time applications*
**6.3 Large Language Models (LLMs)** — *Architecture, training dynamics, fine-tuning, deployment*
**6.4 Generative AI** — *RLHF, diffusion models, full-stack GenAI engineering*
**6.5 RAG — Retrieval-Augmented Generation** — *Building, evaluating, and optimising RAG pipelines*
**6.6 Vector Databases** — *Embeddings, ANN search, ChromaDB, Weaviate, Pinecone, hybrid search*
**6.7 Agentic AI** — *Agents, tool use, multi-agent systems, LangGraph, AutoGen*
**6.8 Prompt Engineering + Fine-Tuning** — *Prompt design, LoRA, QLoRA, PEFT, instruction tuning*
**6.9 FastAPI for AI Services** — *Building and deploying AI-powered APIs*

# PHASE 7 Advanced Track — Cloud · Data Engineering · Certification

📌 Complete Phases 5 and 6 before starting Phase 7. These topics require full deep learning and AI engineering knowledge to use meaningfully.

## 7.1 — Data Engineering

→ **Data Engineering Zoomcamp** — *DataTalksClub · free · Kafka, Spark, dbt, BigQuery, Airflow*
https://datatalks.club/blog/data-engineering-zoomcamp.html

## 7.2 — Google Cloud + ML on Cloud

→ **Machine Learning Path** — *Google Cloud Skills Boost · official & free · Vertex AI, BigQuery ML, Pipelines*
https://www.cloudskillsboost.google/paths/17

## 7.3 — Advanced MLOps at Scale

→ **MLOps Zoomcamp** — *DataTalksClub · complete if not done in Phase 4*
https://datatalks.club/blog/mlops-zoomcamp.html

## 7.4 — Google Professional Machine Learning Engineer ★ Capstone

The Google Professional ML Engineer is one of the most respected cloud ML credentials globally. Attempt only after completing all phases above.

**Step 1** **Google Cloud Skills Boost — ML Path** — *Official pre-certification curriculum — complete first*
https://www.cloudskillsboost.google/paths/17

**Step 2** **Official Exam Guide** — *Free · everything testable is listed here*
https://cloud.google.com/learn/certification/machine-learning-engineer

**Step 3** **Preparing for the Professional ML Engineer Exam** — *Google + Coursera · audit free · official exam prep*
https://www.coursera.org/learn/preparing-cloud-professional-ml-engineer-exam

**Step 4** **Hands-on Vertex AI Labs** — *Google Cloud Skills Boost · complete every ML quest*
https://www.cloudskillsboost.google/

GeeksForGeeks Course (all GFG phases): https://www.geeksforgeeks.org/batch/ds-16?tab=Resources

DeepLearning.AI Short Courses (Phase 6 permanent resource): deeplearning.ai/short-courses

GPU Access — Kaggle Notebooks (Free T4 GPU · 30hrs/week · use for Phase 5, 6.8, and ARENA): www.kaggle.com/code

GPU Access — Google Colab (Free T4 GPU backup · use when Kaggle quota runs out · ): https://colab.research.google.com/